

Towards Multi-Label Text Interpretation with Chain-of-Thought Prompting and Contextualized Knowledge

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Abstract

Existing multi-label topic models face several challenges when interpreting texts annotated with multiple labels: (1) they often associate irrelevant text segments with incorrect labels, which negatively impacts both segment and label interpretation; (2) they fail to effectively capture the semantic relationships between tokens and labels within the segment; (3) they do not integrate contextualized knowledge that could improve interpretability. To overcome these issues, we introduce the Contextualized Prompting Topic Model (CPTM). CPTM utilizes Chain-of-Thought (CoT) prompting to better align text segments with their semantically relevant labels. Furthermore, it integrates label-specific token visualization and topic mining procedure to facilitate the interpretation of tokens and labels. Experimental evaluations conducted on three multi-label text datasets show that CPTM significantly outperforms existing models in both segment and label interpretation. Human assessments also verify CPTM's effectiveness in accurately identifying label-relevant tokens within segments and providing insightful token-level interpretation.

CCS Concepts

• Information systems → Document topic models.

Keywords

Topic Alignment, Multi-Label Topic Modeling, Chain-of-Thought Prompting, Text Interpretation

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1 Introduction

Textual data often convey multiple thematic concepts and may be annotated with several labels that reflect their semantics. For example, the summary of the movie *'Blindpassasjer'* in Fig. 1 is labeled with *'Mystery'*, *'Sci-Fi'*, and *'Thriller'*. Such multi-label texts often encounter the Credit Attribution problem [23], where multiple labels do not always apply with equal specificity across the text, pointing to a critical need for novel approaches to align segments/tokens to the interpretable labels.

To address this challenge, multi-label text interpretation should meet the following requirements: 1). *Segment Interpretation*: Each segment should be aligned with the corresponding labels associated with the text, as shown by the colored underlines in Fig. 1 (top). 2). *Token Interpretation*: The semantic differences between tokens and their aligned labels within each segment should be captured, as indicated by the intensity of highlighted words in Fig. 1 (top). 3). *Label Interpretation*: A semantic interpretation should be provided for labels in the label set, particularly when labels are semantically similar and difficult to distinguish, such as *'Horror'*, *'Mystery'*, and *'Thriller'* in Fig. 1 (bottom).

To interpret text in multi-label scenarios, multi-label topic modeling [1, 7, 35] offers a promising direction and has seen significant progress. Ramage et al. [23] propose the Labeled-LDA which learns label-word correspondence by establishing a one-to-one mapping between latent topics and labels. Similarly, Wang et al. [32] extend the Labeled-LDA by integrating an autoencoding framework, resulting in a neural-based approach called Neural Labeled LDA.

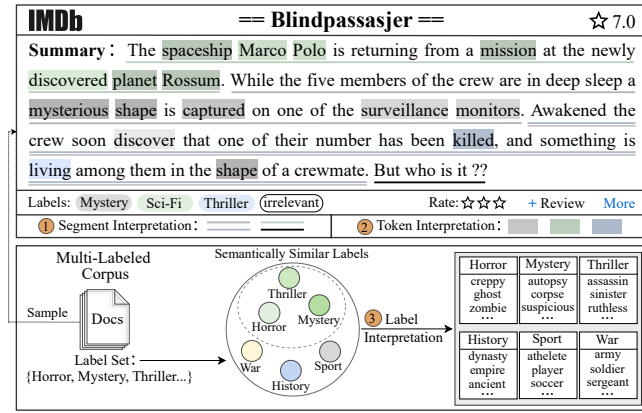


Figure 1: Segment and token interpretations of movie summary ‘Blindpassasjer’, colored underlines indicate labels of segments, and the intensity of highlighted tokens means correlations between tokens and assigned labels. Topics in the box provide label interpretation.

However, these approaches have limitations in the following aspects: 1). They assign tokens from irrelevant segments (highlighted with a black underline in Fig. 1) to predefined labels, thereby affecting the performance of segment and label interpretations. 2). They fail to capture semantic correlations between tokens and labels within segments, hindering token interpretation. 3). They do not integrate external semantic knowledge, particularly contextualized knowledge [25] and large language models (LLMs) [15, 41], to improve interpretation.

To address these limitations, we propose the novel Contextualized Prompting Topic Model (CPTM), which combines the Chain-of-Thought (CoT) prompting [33] and a contextualized transformer [25] for multi-label text interpretation. Specifically, to align each segment¹ with semantically relevant labels and the ‘irrelevant’ label (as shown in Fig. 1), CPTM introduces a CoT reasoning mechanism that integrates keywords, local context, and label descriptions with an LLM for accurate topic alignment. It also constructs the label-specific topic representations using a transformer [25]. Using these topic representations, it then devises a label-specific token visualization scheme to mine and visualize the semantic correspondence between tokens and their associated labels for each segment. Additionally, it generates topics that are semantically consistent with labels in the label set using the label-specific topic mining scheme. By incorporating the ‘irrelevant’ label and contextualized knowledge in LLM [28] and transformer [25], CPTM enhances multi-label text interpretation at the segment, token and label levels jointly.

The main contributions of this work are:

- We propose the novel Contextualized Prompting Topic Model (CPTM), which is the first attempt to combine advanced prompt engineering and contextualized knowledge for multi-label text interpretation.
- CPTM employs the CoT prompting to integrate keywords, local context, and label descriptions with an LLM, aligning segments with semantically relevant labels for segment

interpretation. Additionally, it performs label and token interpretations using a transformer-based language model.

- Experimental results on three publicly available multi-label text corpora demonstrate that CPTM outperforms state-of-the-art baselines in segment and label-level interpretations. Furthermore, human evaluation results confirm CPTM’s effectiveness in highlighting label-relevant tokens within segments and providing meaningful token interpretation.

2 Related Work

Our work is related to three lines of research: multi-label topic modeling, contextualized language models and LLM-based prompt engineering.

2.1 Multi-label Topic Modeling

Multi-label topic modeling [7], which bridges topic modeling and texts with multiple annotations, provides a direction to multi-label text interpretation.

Several attempts have been made, Ramage et al. [23] built a one-to-one correspondence between labels and topics and proposed the Labeled-LDA to address the Credit Attribution problem. Following this, Rubin et al. [27] extended the one-to-one assumption in Labeled-LDA and proposed the Dependency-LDA which models the topic as a mixture of labels to capture the dependency between topics and labels. Similarly, Wang et al. [32] modeled topics with an autoencoding framework and proposed the Neural Labeled LDA. However, these approaches do not leverage semantic knowledge (such as LLM [39] and Contextualized transformer [25]) to improve topic quality.

2.2 Contextualized Language Models

Contextualized language models, such as the BERT [11] and the FLAN-T5 [10], have been explored in various Natural Language Processing (NLP) tasks due to their effectiveness in capturing contextualized linguistic information.

Concretely, Patel and Domeniconi [20] incorporated the subword attention and post-processing mechanism to propose the SPRUCE for generating high-quality sentence embedding. Hawasly et al. [14] explored combining contextualized word representations and clustering algorithms for latent concept discovery. Meanwhile, Meng et al. [18] proposed the TopClus, which is a latent space learning and clustering framework built upon contextualized embeddings, for topic extraction. Similarly, Fang et al. [12] incorporated the contextualized word representations into the topic modeling and proposed the Contextualized Word Topic Model (CWTM).

2.3 LLM-based Prompt Engineering

Advances in LLMs have made them a thriving research topic and have been explored for their robust capabilities in NLP tasks [29, 30, 40].

Meanwhile, Prompt Engineering [16, 28] has emerged as a crucial approach to extend the capability of LLMs. Radford et al. [22] first introduced Zero-shot prompting, which marked a paradigm shift in leveraging LLMs. By providing several demonstrations to induce the target of the given task, Brown [5] proposed the Few-shot

¹In this work, the document is segmented by sentence.



As shown in Fig. 2, our proposed CPTM consists of three components: 1). Topic Alignment and Representation: This module first aligns each segment s_i ($i \in \{1, 2, \dots, M\}$) in document d with the semantically relevant labels in the augmented document-specific label set $L_d^* = L_d \cup \text{'irrelevant'}$, forming the segment-specific label set $L_{s_i} = \{l_1^{s_i}, l_2^{s_i}, \dots, l_{K_{s_i}}^{s_i}\}^2$, using a CoT prompting mechanism. Moreover, it also constructs the label-specific topic representation $\tilde{e}_k^t$ for the k -th topic ($k \in \{1, 2, \dots, K\}$), thereby forming the topic representation matrix E_t . This matrix is then used for token visualization and topic mining with the transformer language model $\mathcal{T}$. 2). Label-Specific Token Visualization: For each segment and its

corresponding segment-specific label set pair $\langle s_i, L_{s_i} \rangle$, this module computes the semantic similarities between tokens and labels using the transformer $\mathcal{T}$ and highlights the label-relevant tokens in the segment. 3). Label-Specific Topic Mining: This module uses the contextualized word representations generated by $\mathcal{T}$ to mine topics that are semantically consistent with the labels. Detailed descriptions and functionalities of these components will be presented individually. For ease of illustration, we list notations that appeared in this paper in Table 1.

4.1 Topic Alignment and Representation

To address the credit attribution problem [23] and enable segment interpretation in multi-label text, CPTM first aligns each segment with semantically relevant labels using a CoT prompting mechanism that integrates the keywords information, local context, and label descriptions generated by an LLM into the reasoning process, as shown in the left panel of Fig. 2.

Concretely, for each document $d = (s_1, s_2, \dots, s_M)$ associated with the label set $L_d = \{l_1^d, l_2^d, \dots, l_{K_d}^d\}$, we first add an 'irrelevant' label to align irrelevant segments and form the associated augmented label set L_d^* . Additionally, we utilize an LLM to generate label descriptions to enrich the semantic information of document-associated labels.

Then, for each segment s_i ($i \in \{1, 2, \dots, M\}$) in the document, we devise a three-step Chain-of-Thought reasoning mechanism to align the topic accurately, as shown in Fig. 3. Similar to human reasoning, CPTM first summarizes the segment's content and identifies keywords, forming a keywords set such as {'detective', 'sharing', 'colleague', 'moment', 'connection'}. Secondly, it filters out irrelevant keywords based on the local context, resulting in a refined keywords set {'sharing', 'moment', 'connection'}. Finally, CPTM aligns the segment with the 'Romance' topic by comparing the semantic correlations between the words in the refined keywords set and the label descriptions. The detailed prompt of this three-step reasoning is shown in Fig. 4, which follows the OpenAI released API³.

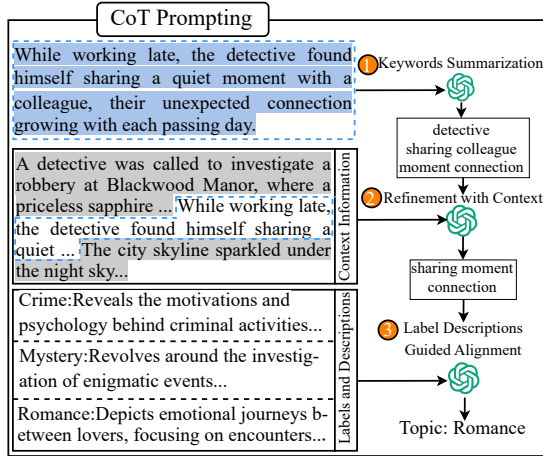


Figure 3: CoT reasoning for topic alignment.

Besides, with these LLM-generated <segment: label set> pairs, we could collect all the segments aligned with label $l_k \in L$ and build a label-specific segment collection D_k for topic representation construction.

Specifically, for the j -th segment $s^j = [w_1^j, w_2^j, \dots, w_{N_{s_j}}^j]$ in collection D_k , we could generate its contextualized representation $\vec{e}^j$ with:

$$[\vec{e}_1^j, \vec{e}_2^j, \dots, \vec{e}_{N_{s_j}}^j] = \mathcal{T}([w_1^j, w_2^j, \dots, w_{N_{s_j}}^j]) \quad (1)$$

$$\vec{e}^j = \frac{1}{N_{s_j}} \sum_{n=1}^{N_{s_j}} \vec{e}_n^j \quad (2)$$

³<https://cookbook.openai.com/examples/>

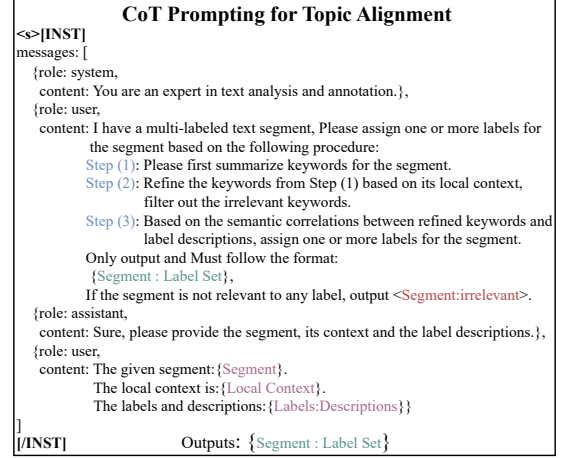


Figure 4: Details of Chain-of-Thought prompting.

where $\mathcal{T}(\cdot)$ is the transformer [25], N_{s_j} means the number of words in segment s^j , $\vec{e}_n^j$ is the contextualized word representation of the n -th word in s^j .

Thus, the topic representation $\vec{e}_k^t$ corresponding to the k -th label l_k could be calculated with:

$$\vec{e}_k^t = \frac{1}{N_k} \sum_{j=1}^{N_k} \vec{e}^j \quad (3)$$

where N_k means the number of segments in D_k . Similarly, the topic representation matrix $E_t \in \mathbb{R}^{H_T \times K}$ can be constructed and used for token visualization and topic mining. Here, H_T means the dimension of transformer embedding.

4.2 Label-Specific Token Visualization

To identify tokens relevant to aligned labels in a segment and visualize the differences in semantic correlations between tokens and labels, we propose a label-specific token visualization scheme using contextualized word representations from the transformer [25], as shown in the top-right panel of Fig. 2.

In detail, given the i -th segment s_i in document d that contains word sequence $[w'_1, w'_2, \dots, w'_{N_{s_i}}]$ and the LLM-responded segment-specific label set $L_{s_i} = \{l_1^{s_i}, l_2^{s_i}, \dots, l_{K_{s_i}}^{s_i}\}$, we first compute the contextualized word representations with transformer $\mathcal{T}$:

$$[\vec{e}'_1, \vec{e}'_2, \dots, \vec{e}'_{N_{s_i}}] = \mathcal{T}([w'_1, w'_2, \dots, w'_{N_{s_i}}]) \quad (4)$$

where $\vec{e}'_n$ means the contextualized word representation of w'_n ($n \in \{1, 2, \dots, N_{s_i}\}$), N_{s_i} represents the number of words in segment s_i , K_{s_i} indicates the number of labels associated with s_i . Then, the semantic correlations between word w'_n and the k' -th label $l_{k'}^{s_i}$ in L_{s_i} could be calculated with:

$$c_{k'}^n = \cos(\vec{e}'_n, \vec{e}_{k'}^{t_{s_i}}) \quad (5)$$

where $\vec{e}_{k'}^{t_{s_i}}$ represents the topic representation, computed using Eq. 3, corresponding to the label $l_{k'}^{s_i}$. Likewise, we could construct the label-token correlation matrix $C_{s_i} \in \mathbb{R}^{N_{s_i} \times K_{s_i}}$ to capture the correlations between tokens in segment s_i and labels in L_{s_i} , which can then be used to identify and visualize label-relevant tokens (with stopwords omitted in the visualization).

4.3 Label-Specific Topic Mining

Aiming at mining label-specific topics to provide label interpretation for each label l_k in the label set $L = \{l_1, l_2, \dots, l_K\}$, we devise a label-specific topic mining scheme based on the transformer $\mathcal{T}$ and label-specific topic representations, as shown in the bottom-right panel of Fig. 2.

Concretely, for the v -th word w_v in the vocabulary ($v \in \{1, 2, \dots, V\}$), we should first collect all its occurrences in the corpus, represented by $W_v = (w_v^1, w_v^2, \dots, w_v^{N_v})$. Then, for the n -th ($n \in \{1, 2, \dots, N_v\}$) occurrence of w_v , denoted with w_v^n , its contextualized word representation $\vec{e}_v^n$ could be obtained with:

$$\vec{e}_v^n = \mathcal{T}(\text{context}(w_v^n)) \quad (6)$$

where $\text{context}(w_v^n)$ represents the specific context of its n -th occurrence, N_v means the number of times w_v appeared in the corpus. And the local word-topic distribution $\vec{c}_v^n$ of w_v^n could be calculated with formulation:

$$\vec{c}_v^n = \text{smax}([s(\vec{e}_v^n, \vec{e}_1^t), s(\vec{e}_v^n, \vec{e}_2^t), \dots, s(\vec{e}_v^n, \vec{e}_K^t)]) \quad (7)$$

where $s(\cdot, \cdot)$ means the cosine similarity, $\text{smax}(\cdot)$ is the softmax normalization, $\vec{e}_k^t$ ($k \in \{1, 2, \dots, K\}$) denotes the topic representation corresponding to the k -th label.

Thus, the corpus-level word-topic distribution $\vec{c}_v$ of w_v could be obtained using:

$$\vec{c}_v = \frac{1}{N_v} \sum_{n=1}^{N_v} \vec{c}_v^n \quad (8)$$

Similarly, we could calculate corpus-level word-topic distributions for all the words in the vocabulary and construct the word-topic distribution matrix $C' \in \mathbb{R}^{V \times K}$. Finally, the topic-word distributions of label-specific topics $\Phi \in \mathbb{R}^{V \times K}$ could be obtained by performing column-wise normalization on C' with:

$$\Phi = \text{norm}_c(C') \quad (9)$$

where $\text{norm}_c(\cdot)$ is the column-wise normalization, the k -th column $\vec{\phi}_k$ represents the word distribution corresponding to the k -th label in the label set L .

5 Experiments

5.1 Experimental Setups

Datasets: To validate the effectiveness of our proposed CPTM, we utilize three publicly available multi-label datasets (IMDB ⁴, UK-LEX ⁵ and EUR-LEX ⁶). Detailed descriptions are summarized as: IMDB [24] contains plot and label information of approximately 117k movies, where each document is annotated with multiple genre labels such as ‘sport’, ‘romance’, and ‘history’. UK-LEX [9] is constituted by legal documents from the United Kingdom’s National Archives ⁷, categorized into topic labels including ‘finance’, ‘transportation’ and ‘education’. EUR-LEX [8] is a collection of documents containing European Union (EU) legislation, annotated with conceptual labels by the EU’s Publications Office. In our experiments, we select English documents tagged with the top 100 most frequent labels.

⁴<https://github.com/shashankvmaiya/Movie-Genre-Multi-Label-Text-Classification>

⁵https://drive.google.com/file/d/1HNH2yTvr44_HtNI9XurzOGWV-PWUNDi/view?usp=share_link

⁶<https://github.com/lancopku/SGM>

⁷<https://www.legislation.gov.uk/>

For all datasets, we conduct a series of pre-processing, such as stopwords, punctuation and non-English token removal, lemmatization and spell-checking with NLTK ⁸ and Spacy ⁹ packages. We also filter out documents with fewer than 3 segments. The statistics of the processed datasets are listed in Table 2 ¹⁰

Table 2: The statistics of the processed datasets.

Dataset	#Docs	#Vocabs	#Avg Length	#Labels	#Avg Labels
IMDB	85,477	5,984	23.6	27	1.99
UK-LEX	30,334	5,728	90.3	40	1.47
EUR-LEX	62,133	6,675	93.9	100	4.51

Baselines: We choose the following state-of-the-art unsupervised (•) and multi-label (*) topic models as baselines: • **LDA** [3], • **CTMNeg** [2], • **BERTopic** [13], • **vONT** [37], • **ECRTM** [34], • **FASTopic** [36], • **CWTM** [12], • **NeuroMax** [21], • **LLM-TM** [19], * **Labeled-LDA** [23], * **Fast Dep.-LLDA** [6], * **Dependency-LDA** [27]. Details of baselines, please refer to Appendix A. For all the baselines, we utilize the officially released code in the experiments. Meanwhile, we follow the default parameter configurations for these approaches. We alias Labeled-LDA, Dependency-LDA and Fast Dep.-LLDA as L-LDA, D-LDA and F-LDA respectively for simplicity in the experiments. For the implementation details, please refer to Appendix B.

Evaluation Metrics: Our evaluation encompasses three levels of interpretation: label-level, segment-level, and token-level analysis. For the label interpretation evaluation, we select four widely used topic coherence metrics (CP, CA, NPML, and UCI) [26], calculated with the released Palmetto ¹¹ library, along with a Unique Term Rate (UTR) [31] metric to assess topic diversity. To evaluate the semantic relatedness between extracted topics and labels, we formulate two metrics based on OpenAI embeddings ¹². The Label Discovered Rate (LDR) measures the proportion of labels that are semantically captured by the extracted topics:

$$\text{LDR} = \frac{1}{K} \sum_{k=1}^K \max(0, \text{sgn}(\cos^*(l_k, \Phi) - \sigma)) \quad (10)$$

where σ is the threshold hyper-parameter, $\text{sgn}(\cdot)$ means the sign function which outputs 1 with positive inputs, $\cos^*(l_k, \Phi)$ represents the maximum cosine similarity between the k -th label l_k and topics. The Correlation Score (CS) quantifies the semantic similarity between labels and their corresponding topics:

$$\text{CS} = \frac{1}{K} \sum_{k=1}^K \cos(l_k, \vec{\phi}_k^{10}) \quad (11)$$

where l_k denotes the k -th label, $\vec{\phi}_k^{10}$ represents the top ten words in the word distribution of the k -th topic.

On the other hand, for segment interpretation evaluation, we adopt the Precision, Recall, MacroF1 and MicroF1 [23] metrics.

The token interpretation evaluation employs Kendall’s Tau correlation τ [4] and Spearman’s rank correlation ρ [17] to assess the agreements of the ranking results of label-related tokens provided

⁸<https://www.nltk.org/>

⁹<https://spacy.io/>

¹⁰For contextualized approaches, we utilize the original data with sequence information in the experiment.

¹¹<https://github.com/dice-group/Palmetto>

¹²<https://platform.openai.com/docs/guides/embeddings>

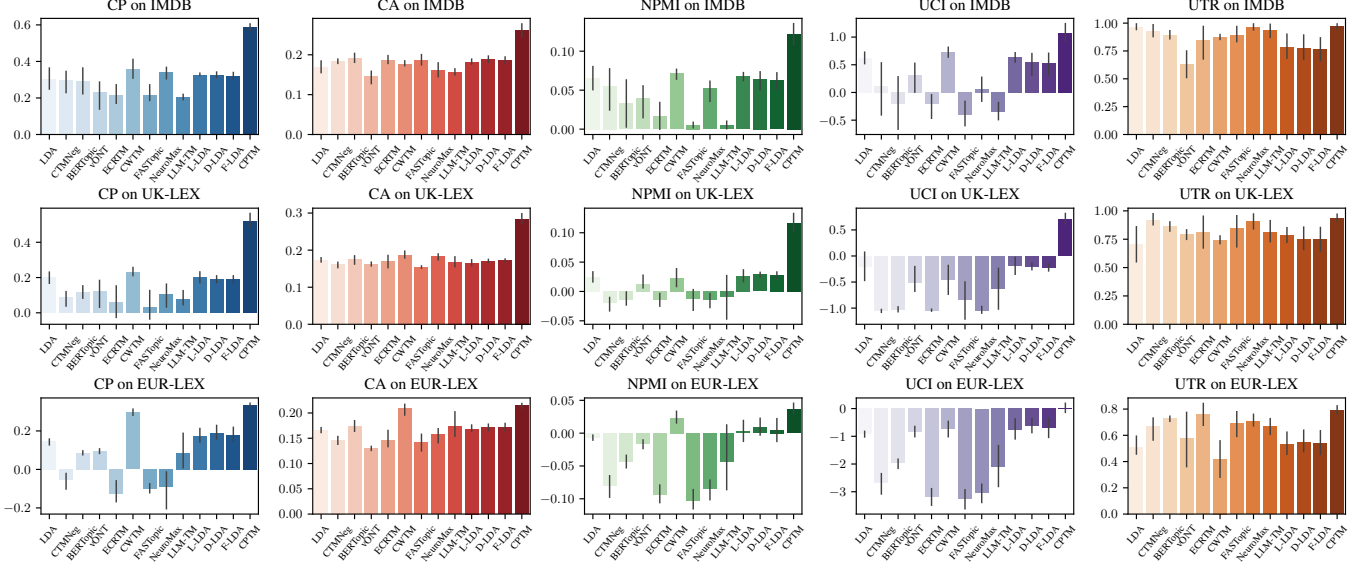


Figure 5: Detailed comparison on coherence and diversity metrics, the height of colored bars means the average value over four topic number settings, and the length of error bars (black short lines) denotes a 95% confidence interval and indicates the statistical variability.

by human annotators and the proposed CPTM. Specifically, the Kendall's Tau correlation coefficient τ is calculated with:

$$\tau = \frac{2(P - Q)}{n(n - 1)} \quad (12)$$

where P and Q denote the number of concordant and discordant instances among two ranked lists, n represents the number of instances in the ranked list. Similarly, the Spearman's Rank Correlation coefficient ρ follows definition:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (13)$$

where d_i means the relative distance of the i -th instance between two ranked lists, and n is the number of instances in the list.

5.2 Label Interpretation Evaluation

To assess the performance of label interpretation, we consider both topic quality and semantic correlations between extracted topics and labels.

Topic Quality: We first perform the topic quality evaluation with four topic coherence (CP, CA, NPMI and UCI) and topic diversity (UTR) metrics. In detail, we set the topic number to four different values¹³ for each dataset and utilize documents associated with the selected labels to validate the effectiveness and stability of CPTM. The numerical comparison results are provided in Table 3. Each value is calculated by averaging the coherence and UTR values across the four topic number settings, with optimal results highlighted in bold. The last column (Rank) presents the overall ranking based on five metrics. Additionally, we present experimental results and error bars in Fig. 5. These results show that our proposed CPTM outperforms competitive baselines in coherence and diversity metrics while maintaining shorter error bars, indicating greater

Table 3: The numerical comparison of topic coherence and topic diversity on three datasets.

Dataset	Model	CP	CA	NPMI	UCI	UTR	Rank
IMDB	LDA	0.302	0.170	0.065	0.616	0.968	3
	CTMNeg	0.297	0.183	0.055	0.123	0.933	7
	BERTopic	0.293	0.192	0.033	-0.200	0.898	10
	vONT	0.233	0.147	0.039	0.320	0.631	9
	ECRTM	0.217	0.187	0.016	-0.203	0.852	11
	CWTM	0.358	0.178	0.071	0.726	0.877	2
	FASTopic	0.219	0.188	0.005	-0.404	0.891	13
	NeuroMax	0.342	0.161	0.052	0.059	0.968	8
	LLM-TM	0.207	0.157	0.005	-0.353	0.940	12
	L-LDA	0.328	0.183	0.067	0.633	0.787	4
	D-LDA	0.328	0.188	0.064	0.552	0.773	5
	F-LDA	0.322	0.187	0.063	0.533	0.768	6
	CPTM	0.587	0.261	0.122	1.143	0.973	1
UK-LEX	LDA	0.199	0.174	0.025	-0.198	0.705	5
	CTMNeg	0.088	0.161	-0.020	-1.048	0.917	12
	BERTopic	0.114	0.174	-0.014	-1.023	0.862	11
	vONT	0.122	0.163	0.013	-0.506	0.796	7
	ECRTM	0.058	0.170	-0.014	-1.039	0.813	13
	CWTM	0.229	0.187	0.024	-0.443	0.738	6
	FASTopic	0.032	0.154	-0.012	-0.830	0.845	9
	NeuroMax	0.105	0.182	-0.014	-1.039	0.907	10
	LLM-TM	0.077	0.169	-0.010	-0.628	0.812	8
	L-LDA	0.201	0.166	0.027	-0.189	0.785	2
	D-LDA	0.192	0.171	0.029	-0.206	0.751	3
	F-LDA	0.188	0.172	0.028	-0.219	0.748	4
	CPTM	0.520	0.284	0.116	0.709	0.933	1
EUR-LEX	LDA	0.146	0.166	-0.006	-0.903	0.506	7
	CTMNeg	-0.057	0.147	-0.080	-2.678	0.664	10
	BERTopic	0.084	0.173	-0.043	-1.973	0.726	8
	vONT	0.095	0.130	-0.016	-0.832	0.580	6
	ECRTM	-0.128	0.146	-0.094	-3.191	0.758	12
	CWTM	0.296	0.208	0.023	-0.739	0.419	3
	FASTopic	-0.099	0.142	-0.103	-3.272	0.690	13
	NeuroMax	-0.093	0.157	-0.084	-3.027	0.707	11
	LLM-TM	0.084	0.173	-0.043	-2.107	0.664	9
	L-LDA	0.171	0.168	0.003	-0.759	0.530	5
	D-LDA	0.191	0.171	0.009	-0.614	0.547	2
	F-LDA	0.180	0.171	0.005	-0.714	0.540	4
	CPTM	0.336	0.214	0.036	0.030	0.787	1

¹³Topic numbers are set to [7, 14, 21, 27], [10, 20, 30, 40] and [25, 50, 75, 100] for IMDB, UK-LEX and EUR-LEX.

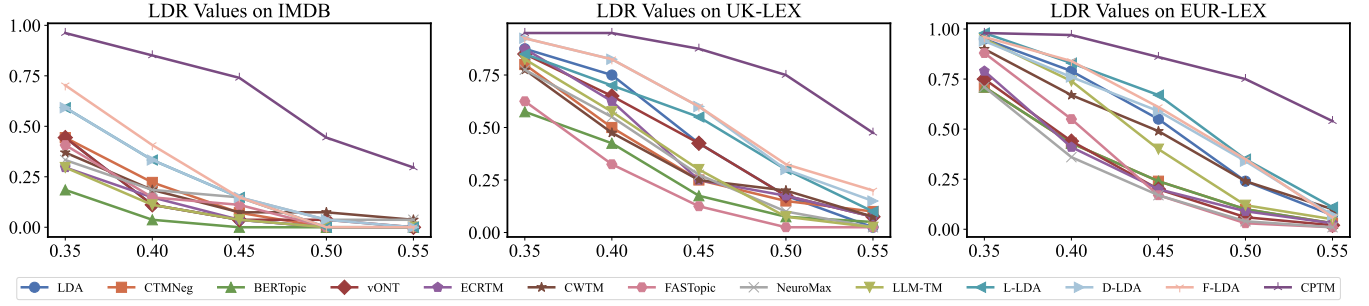


Figure 6: The comparison of LDR metric on IMDB, UK-LEX and EUR-LEX with various threshold parameters σ .

stability across different topic numbers. This improvement in label interpretation can be attributed to two key factors: 1). CoT prompting-based topic alignment helps identify the correct semantic topics for the segments. 2). The incorporation of contextualized knowledge during the topic-mining process enhances topic quality. **Semantic Correlations:** Beyond the topic quality evaluation, we also examine the semantic correlations between extracted topics and labels. We compute the Label Discovered Rate (LDR) values using Eq. 10 with threshold hyper-parameter σ set to [0.35, 0.4, 0.45, 0.5, 0.55] on full datasets. As shown in Fig. 6, our proposed CPTM obtains more topics that share higher semantical similarity with labels in the label set, indicating superiority in the LDR metric. Additionally, with the increase in σ , CPTM’s values decrease more gradually than the baselines, demonstrating its robustness in uncovering label-specific topics.

Table 4: The CS values comparison on three datasets.

Dataset	L-LDA	D-LDA	F-LDA	CPTM
IMDB	0.308	0.342	0.337	0.496
UK-LEX	0.413	0.458	0.450	0.534
EUR-LEX	0.405	0.389	0.406	0.529

Likewise, we also calculate the Correlation Score with Eq. 11 and present the values in Table 4. Here, topic numbers are set to 27, 40 and 100 for the IMDB, UK-LEX and EUR-LEX, unsupervised baselines are omitted as their extracted topics do not correspond to the specific labels. It can be observed that CPTM surpasses the compared baselines by a large margin, which can be attributed to the accurate topic alignment conducted through the Chain-of-Thought Prompting. Detailed ablation study refers to Appendix C. We also provide topic examples in Appendix D to validate the effectiveness of CPTM in mining coherent and label-specific topics intuitively.

5.3 Segment Interpretation Evaluation

A key component of our framework is the LLM-based topic alignment mechanism, which forms the foundation for multi-level interpretation. To evaluate its effectiveness, we conduct experiments on manually annotated test sets. Specifically, we randomly sample three subsets from IMDB/UK-LEX/EUR-LEX datasets and manually annotate each segment with its most appropriate label from the document-specific label set, resulting in 613 (IMDB), 803 (UK-LEX), and 996 (EUR-LEX) annotated <Segment: Label> pairs. Following [23], we employ Precision, Recall, MacroF1, and MicroF1 metrics for evaluation.

Table 5: Topic alignment comparison on three datasets.

Dataset	Model	#Precision	#Recall	%MacroF1	%MicroF1
IMDB	L-LDA	0.468	0.483	0.450	0.492
	D-LDA	0.622	0.593	0.561	0.603
	F-LDA	0.454	0.261	0.223	0.267
	CPTM	0.768	0.817	0.743	0.804
UK-LEX	L-LDA	0.518	0.543	0.489	0.553
	D-LDA	0.685	0.612	0.586	0.628
	F-LDA	0.675	0.589	0.547	0.572
	CPTM	0.796	0.805	0.785	0.832
EUR-LEX	L-LDA	0.464	0.400	0.389	0.399
	D-LDA	0.528	0.451	0.450	0.458
	F-LDA	0.469	0.262	0.243	0.268
	CPTM	0.717	0.629	0.611	0.651

The experimental results in Table 5 demonstrate the superior performance of CPTM in segment-level topic alignment compared with multi-labeled approaches. Several factors may contribute to this improvement: 1). The keywords summarization mechanism effectively filters out noise tokens, allowing CPTM to focus on semantically relevant content. 2). The contextual information integration helps calibrate the semantic scope of segments, leading to more accurate topic alignment. 3). The incorporation of label descriptions enriches the semantic representations beyond the limited information contained in label names alone. These results confirm that our LLM-based topic alignment mechanism effectively captures the semantic relationships between segments and labels.

5.4 Token Interpretation Evaluation

Token interpretation aims to identify semantic-relevant tokens within the segment for each label associated with the segment. To validate the effectiveness of CPTM in token interpretation, we perform human evaluations.

Firstly, we randomly select 65 <Segment: Label>¹⁴ pairs for each dataset. For each pair, we select the top five tokens, generated by CPTM, that are semantically relevant to the associated label as ranking candidates. Then, we recruit two expert annotators to rank the candidate tokens according to the semantic similarities between the tokens and the label. To measure the agreements between ranking results provided by human annotators and CPTM, we leverage two ranking metrics: Kendall’s Tau Correlation [4] and Spearman’s Rank Correlation [17]. The results are presented

¹⁴We select segments assigned with one label for simplicity.

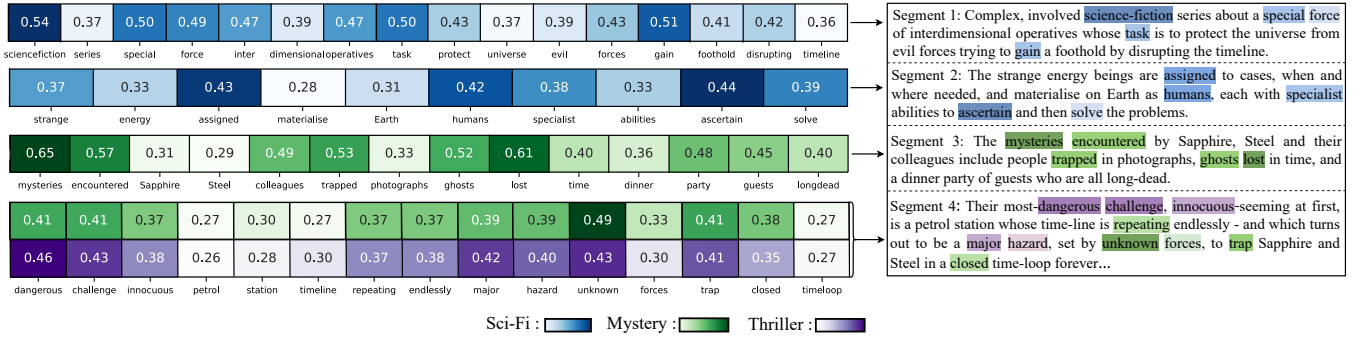


Figure 7: Segment and label-specific tokens visualizations of CPTM on movie *Sapphire & Steel* plot summary.

in Table 6. ‘Human-Only’ indicates the agreement between annotators, ‘Human & CPTM’ shows the agreement between annotators and CPTM. These statistics reflect the strong agreement between human judgments and CPTM’s ranking results. Details of human evaluations please refer to Appendix E.

Table 6: The measured agreement with Kendall’s Tau Correlation and Spearman’s Rank Correlation.

Options	Metrics	IMDB	UK-LEX	EUR-LEX
Human-Only	Kendall τ	0.553	0.798	0.733
	Spearman ρ	0.604	0.859	0.786
Human & CPTM	Kendall τ	0.689	0.704	0.716
	Spearman ρ	0.725	0.779	0.762

5.5 Visualization

To provide an intuitive insight into the multi-level interpretation capability of CPTM, we visualize segment-level, token-level, and label-level analyses. Fig. 7 illustrates the segment and token interpretations of the movie plot summary “Sapphire & Steel” in the IMDB dataset. At the segment level, CPTM identifies distinct topical transitions: the initial segments are associated with ‘Sci-Fi’, followed by a segment related to ‘Mystery’, and concluding with a segment that exhibits dual topics of ‘Mystery’ and ‘Thriller’. On the right of Fig. 7, the token-level visualization reveals the semantic relationships between individual tokens and their assigned labels. The top-5 most significant tokens are highlighted in the segments and stopwords are excluded for clarity.

At the label level, Fig. 8 presents representative topics extracted by CPTM and LDA for corresponding label categories, revealing several key advantages of our approach. While LDA struggles to maintain semantic distinctions between related categories (e.g., ‘Sport’ and ‘Game-Show’), CPTM successfully preserves these fine-grained topical boundaries. Furthermore, CPTM demonstrates better topic discovery capabilities, identifying topics like ‘Mystery’ that LDA fails to capture. This can be attributed to two key factors: 1). The integration of label signals through CoT prompting for label-topic correspondence. 2). The incorporation of contextualized knowledge in the mining process, which contributes to higher topic quality.

6 Conclusion

In this paper, we have proposed the Contextualized Prompting Topic Model (CPTM) for interpreting multi-label text corpora at

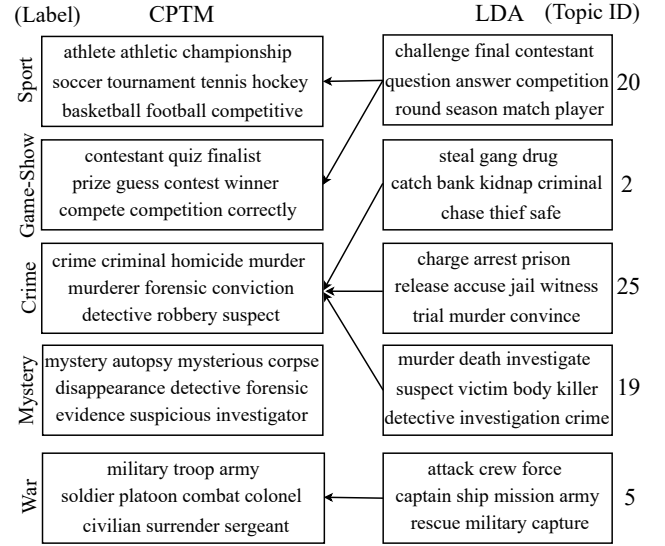


Figure 8: Topic examples comparison with LDA.

segment, token and label levels. CPTM integrates keywords, local context, and label description information with a large language model (LLM), and employs a Chain-of-Thought (CoT) prompting for precise topic alignment. Additionally, it leverages semantic knowledge in transformers to visualize label-specific tokens within segments and generate label-specific topics for token and label interpretations. Experimental results on three multi-label text corpora reveal that CPTM outperforms state-of-the-art baselines in segment and label interpretations. Furthermore, human evaluation results further validate the effectiveness of CPTM in identifying label-relevant tokens and providing token-level interpretation.

7 Acknowledgements

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A Baselines

We choose the following state-of-the-art unsupervised (•) and multi-label (*) topic models as baselines:

- **LDA** [3] is the most widely used topic model based on Bayesian statistics ¹⁵.
- **CTMNeg** [2] is a contextualized neural topic model based on contrastive learning and negative sampling ¹⁶.
- **BERTopic** [13] is a topic mining approach based on contextualized representation learning ¹⁷.
- **vONT** [37] is a VAE-based neural topic model which models topics with von Mises-Fisher distribution ¹⁸.
- **ECRTM** [34] is a neural topic model based on the optimal transport ¹⁹.
- **CWTM** [12] is a neural topic model based on pre-trained language model ²⁰.
- **FASTopic** [36] leverages optimal transport between the document, topic, and word embeddings from pre-trained transformers to model topics ²¹.
- **NeuroMax** [21] is a neural topic model based on PLMs and mutual information ²².
- **LLM-TM** [19] is a topic mining approach based on LLM and prompt engineering ²³.
- * **Labeled-LDA** [23] is the first multi-label topic model which aims to mine label-specific topics ²⁴.
- * **Dependency-LDA** [27] is an extension of Labeled-LDA, it models each topic as a mixture of labels to capture the dependency between topics and labels ²⁵.
- * **Fast Dep.-LLDA** [6] extends the Dependency-LDA with a greedy layer-wise training mechanism for acceleration ²⁶.

B Implementation Details

All experiments are conducted on a machine running Ubuntu, equipped with an NVIDIA RTX 3090Ti GPU (CUDA 12.2) and 64GB RAM. In our main experiments, we employ GPT-3.5-Turbo ²⁷ as the LLM for topic alignment and all-mpnet-base-v2 ²⁸ as the PLM for topic mining and token visualization.

C Ablation Study

To further investigate the effectiveness of different components in CPTM, we conduct ablation studies on the IMDB dataset with 7 topics. The experiments focus on four critical aspects: the analysis of CoT prompting strategies, the impact of different LLMs, the

selection of PLMs, and the effectiveness of irrelevant segment handling. For comparison, we focus on CWTM as the representative unsupervised approach and L-LDA as the multi-label approach due to their superior performance in the baselines.

CoT Prompting Analysis: To validate the effectiveness of our prompt, we compare it with the Standard CoT [33] and Hierarchical CoT [38] on topic modeling performance. Also, the prompt in CPTM consists of different procedures and information for topic alignment. To understand the contribution of different prompting components, we ablate our prompting strategy by removing three key elements: keywords summarization, label descriptions, and contextual information from the complete prompt. Table 7 presents the detailed comparison across different configurations. The results demonstrate that our prompt outperforms the existing advanced prompting strategies and each component contributes meaningfully to the performance of topic quality. Notably, the zero-shot setting, which excludes all specialized prompting components, shows the lowest performance. Even in its most basic configuration, CPTM maintains superior performance compared to baseline methods.

Table 7: Comparison of different CoT prompting components on topic modeling performance.

Model & Strategy	CP	CA	NPMI	UCI	UTR	CS
CWTM	0.286	0.171	0.060	0.671	0.885	–
L-LDA	0.322	0.186	0.073	0.771	0.971	0.382
Standard CoT	0.531	0.264	0.125	1.354	0.986	0.467
Hierarchical CoT	0.559	0.277	0.135	1.432	1.000	0.452
Zero-Shot	0.524	0.255	0.121	1.322	0.986	0.472
w/o Keywords	0.542	0.259	0.128	1.414	0.986	0.466
w/o Label Descriptions	0.544	0.266	0.130	1.423	0.986	0.459
w/o Context	0.551	0.263	0.127	1.404	0.986	0.456
w/ all-Information	0.602	0.278	0.140	1.435	1.000	0.481

Impact of Different LLMs: To study the generalizability of our approach across different LLMs, we evaluate CPTM with six popular LLMs: Mistral-Nemo ²⁹, Mistral-Large ³⁰, GPT-3.5-Turbo ³¹, GPT-4o ³², GPT-4o-mini ³³, and GPT-4 ³⁴ and three recently emerged reasoning LLMs: o1 ³⁵, o3-mini ³⁶, and DeepSeek-R1 ³⁷. The detailed results are listed in Table 9. While all CPTM variants outperform the baseline methods in topic quality, GPT-4 achieves the best results across all evaluation criteria. Furthermore, the evaluation statistics indicate that all variants exhibit improved capability in mining topics with higher semantic consistency with labels.

PLM Selection Analysis: We also explore the impact of different PLMs on the performance of CPTM. Table 10 presents results across six widely-adopted PLMs. The experimental results demonstrate that although the numerical statistics fluctuate with different PLM architectures, CPTM consistently outperforms comparable

¹⁵<https://mimno.github.io/Mallet/>

¹⁶<https://github.com/AdhyaSuman/CTMNeg>

¹⁷<https://github.com/MaartenGr/BERTopic>

¹⁸<https://github.com/xuweijieshuai/NeuralTopicModeling-vmf>

¹⁹<https://github.com/bobxwu/ecrtm>

²⁰<https://github.com/Fitz-like-coding/CWTM>

²¹<https://github.com/bobxwu/FASTopic>

²²<https://github.com/Fsoft-AIC/NeuroMax>

²³<https://github.com/GateNLP/LLMs-for-Topic-Modeling>

²⁴<https://github.com/JoeZJH/Labeled-LDA-Python>

²⁵<https://github.com/timothyruhin/DependencyLDA>

²⁶<https://github.com/sophieburkhardt/Multi-Label-Topic-Modeling>

²⁷<https://openai.com/index/gpt-3-5-turbo-fine-tuning-and-api-updates/>

²⁸<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

²⁹<https://mistral.ai/news/mistral-nemo>

³⁰<https://mistral.ai/en/news/mistral-large>

³¹<https://openai.com/index/gpt-3-5-turbo-fine-tuning-and-api-updates/>

³²<https://platform.openai.com/docs/models/gpt-4o>

³³<https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>

³⁴<https://openai.com/index/gpt-4-api-general-availability/>

³⁵<https://openai.com/o1/>

³⁶<https://openai.com/index/openai-o3-mini/>

³⁷<https://api-docs.deepseek.com/news/news250120>

Table 8: Comparison of irrelevant segments ablation on the IMDB dataset, ‘w/’ and ‘w/o’ mean with and without irrelevant segments, Δ columns (Δ_{CP} , Δ_{CA} , Δ_{NPMI} , Δ_{UCI} and Δ_{UTR}) are increments of filtering the irrelevant segments.

Model	CP (w/)	CP (w/o)	Δ_{CP}	CA (w/)	CA (w/o)	Δ_{CA}	NPMI (w/)	NPMI (w/o)	Δ_{NPMI}	UCI (w/)	UCI (w/o)	Δ_{UCI}	UTR (w/)	UTR (w/o)	Δ_{UTR}
LDA	0.245	0.269	+0.024	0.158	0.169	+0.011	0.050	0.061	+0.011	0.510	0.621	+0.111	1.000	1.000	–
CTMNeg	0.192	0.274	+0.082	0.178	0.196	+0.018	0.010	0.029	+0.019	-0.699	-0.479	+0.220	1.000	1.000	–
BERTopic	0.377	0.430	+0.053	0.207	0.253	+0.046	0.075	0.107	+0.032	0.576	1.035	+0.459	0.943	0.957	+0.014
vONT	0.091	0.106	+0.015	0.117	0.122	+0.005	0.001	0.010	+0.009	-0.157	-0.040	+0.117	0.757	0.786	+0.029
ECRTM	0.216	0.299	+0.083	0.183	0.220	+0.037	0.044	0.058	+0.014	-0.019	0.373	+0.392	1.000	1.000	–
CWTM	0.286	0.379	+0.093	0.171	0.204	+0.033	0.060	0.083	+0.023	0.671	0.878	+0.207	0.885	0.957	+0.072
FASTopic	0.299	0.342	+0.043	0.208	0.212	+0.004	0.007	0.077	+0.070	-0.038	0.769	+0.807	1.000	1.000	–
NeuroMax	0.285	0.371	+0.086	0.158	0.239	+0.081	0.061	0.077	+0.016	0.376	0.504	+0.128	0.971	1.000	+0.029
LLM-TM	0.215	0.506	+0.291	0.156	0.238	+0.082	0.006	0.111	+0.105	-0.425	1.059	+1.484	0.957	0.971	+0.014
L-LDA	0.322	0.356	+0.034	0.186	0.202	+0.016	0.073	0.077	+0.004	0.771	0.795	+0.024	0.971	0.986	+0.015
D-LDA	0.338	0.363	+0.025	0.200	0.206	+0.006	0.074	0.085	+0.011	0.667	0.905	+0.238	0.914	0.943	+0.029
F-LDA	0.334	0.385	+0.051	0.197	0.211	+0.014	0.072	0.091	+0.019	0.652	0.974	+0.322	0.914	0.929	+0.015
CPTM	0.570	0.602	+0.032	0.274	0.278	+0.004	0.133	0.140	+0.007	1.323	1.435	+0.112	1.000	1.000	–

Table 9: Comparison of CPTM with different LLMs.

Model	CP	CA	NPMI	UCI	UTR	CS
CWTM	0.286	0.171	0.060	0.671	0.885	–
L-LDA	0.322	0.186	0.073	0.771	0.971	0.382
CPTM(Mistral-Nemo)	0.574	0.267	0.132	1.467	1.000	0.481
CPTM(Mistral-Large)	0.624	0.273	0.140	1.581	1.000	0.481
CPTM(GPT-3.5-Turbo)	0.602	0.278	0.140	1.435	1.000	0.481
CPTM(GPT-4o)	0.613	0.279	0.143	1.603	1.000	0.485
CPTM(GPT-4o-mini)	0.646	0.279	0.141	1.529	1.000	0.480
CPTM(o1)	0.508	0.260	0.120	1.284	1.000	0.466
CPTM(o3-mini)	0.516	0.262	0.120	1.263	1.000	0.486
CPTM(DeepSeek-R1)	0.514	0.241	0.113	1.231	1.000	0.485
CPTM(GPT-4)	0.650	0.287	0.152	1.760	1.000	0.488

approaches. This consistency proves its robustness to the selection of PLMs.

Table 10: Analysis of CPTM performance with various PLM architectures.

Model	CP	CA	NPMI	UCI	UTR	CS
CWTM	0.286	0.171	0.060	0.671	0.885	–
L-LDA	0.322	0.186	0.073	0.771	0.971	0.382
CPTM(bert-base-uncased ³⁸)	0.508	0.245	0.117	1.288	1.000	0.434
CPTM(bert-base-nli-mean-tokens ³⁹)	0.543	0.255	0.117	1.217	1.000	0.421
CPTM(sentenceT5 ⁴⁰)	0.625	0.260	0.139	1.591	1.000	0.490
CPTM(all-mpnet-base-v2 ⁴¹)	0.602	0.278	0.140	1.435	1.000	0.481
CPTM(all-MiniLM-L6-v2 ⁴²)	0.654	0.290	0.141	1.552	1.000	0.484
CPTM(paraphrase-MiniLM-L6-v2 ⁴³)	0.668	0.288	0.151	1.596	1.000	0.480

Irrelevant Segments Analysis: A distinctive assumption of CPTM is its ability to ignore irrelevant segments that are not related to the document-annotated labels through the incorporation of an ‘irrelevant’ label. To validate this design choice, we conduct experiments to observe the impact of the ‘irrelevant’ label for the label interpretation. Concretely, with the feedback from LLMs, we filter out the irrelevant segments in documents and explore the topic quality variation. The comparative analysis in Table 8 reveals consistent improvements in topic quality metrics when irrelevant segments are properly identified and filtered.

D Topic Examples

To demonstrate the effectiveness of CPTM in extracting coherent and label-specific topics, we present the topics extracted by both CPTM and L-LDA from the IMDB dataset, with the number of topics set to 27.

From the topics extracted in Table 11, we could observe the following findings: 1). CPTM generates more coherent topics than

L-LDA. 2). Topics extracted by CPTM exhibit stronger semantic correlations with the labels. 3) CPTM extracts topics that are semantically relevant to all labels, whereas L-LDA fails to identify topics related to ‘Adult’, ‘Animation’, ‘Drama’, ‘Comedy’, ‘Documentary’, ‘Family’, ‘Musical’ and ‘Mystery’. These improvements can be attributed to the following factors: 1). CoT prompting aligns segments with semantically relevant labels more accurately. 2). The transformer-based topic-mining framework integrates semantic knowledge into the modeling process, enhancing the quality of the extracted topics.

E Details of Human Evaluations

To conduct human evaluations on token interpretation, we invited two master’s students majoring in computer science, both of whom are familiar with natural language processing. Both participants are informed that they are conducting human evaluations using an automatic method, and that no private information is collected during the evaluation process. The instructions used in the human evaluation experiments are shown in Fig. 9

Human Evaluation---Token Interpretation

Your task is to assess the semantic relevance of each of the five given words to the label associated with the provided sentence. Then, rank these five words from highest to lowest based on their relevance and fill in the blanks accordingly.

* 1. The given sentence is : {A legendary event in Japan, PRIDE FC matches the world's fiercest fighters in competitions of strength and honor, featuring and intense mix of combative sports such as wrestling, boxing, jiu-jitsu, karate, judo, kickboxing and others.} and its assigned label is : { Sport }

The five-tokens are: {strength intense event match competition}

1.
2.
3.
4.
5.

Submit

Figure 9: Human evaluation example in the token interpretation section.

Table 11: Topic words comparison with L-LDA on IMDB dataset (27 labels), ‘—’ means no matched label.

Model	Labels/Topics	Topic Words
CPTM	Action	action activate shield avenger command attack fury hulk commander defender
	Adult	adult sexual sexually porn nude lesbian pleasure naked sexy kiss
	Adventure	adventure expedition explorer wilderness quest voyage cave treasure exploration island treasure
	Animation	animation animate dinosaur frog mouse balloon puppet bubble dragon mechanical
	Biography	biography historian founder author career profile famous influential history historical
	Comedy	comedic humor comedian parody hilarious funny joke sitcom sketch laugh
	Crime	crime criminal homicide murder murderer forensic conviction detective robbery suspect
	Documentary	documentary filmmaker footage explore wildlife examine researcher archive cultural cinema
	Drama	drama tension bitterly tragedy unhappy dramatic reconcile emotionally affair upset
	Family	parent aunt household grandmother nephew grandma grandpa grandfather cousin niece
	Fantasy	fantasy magical magic fairy realm spell witch wizard sword demon
	Game-Show	contestant quiz finalist prize guess contest winner compete correctly competition
	History	history historical historian dynasty empire ancient historic emperor century revolution
	Horror	horror ghostly scary creepy haunt ghost zombie paranormal nightmare supernatural
	Music	musician song singer album band musical sing vocal tune guitar
	Musical	musical melody sing song theater tune piano vocal musician performance
	Mystery	mystery autopsy mysterious disappearance corpse detective forensic mysteriously suspicious investigator
	News	news newspaper journalist correspondent republican presidential trump democratic headline election
	Reality-TV	reality chef kitchen culinary contestant dish cook menu housewife elimination
	Romance	romance romantic affection relationship admirer kiss lover seduce girlfriend marry
	Sci-Fi	spaceship alien scientist radiation shuttle science planet space scientific astronaut
	Short	length tall simple height monologue size tiny nickname strength easy
	Sport	athlete athletic championship soccer tournament tennis hockey basketball football competitive
	Talk-Show	podcast panelist conversation chat presenter speaker discussion speech speak panel
	Thriller	thriller assassin assassination terrorist murderous plot sinister hostage ruthless kidnapper
	War	military troop army soldier platoon combat colonel surrender civilian sergeant
	Western	western rancher outlaw cattle wagon saloon west railroad ranch cowboy
L-LDA	Action	rescue attack steal force fight mission gang capture arrive secret
	—	hotel gunter kitchen poppy offer space idea sell inform project
	Adventure	island ship captain crew water land rescue travel boat river
	—	dragon monster hero adventure bear steal power fall super book
	Biography	career documentary character explore journey experience viewer actor personal review
	—	parent relationship invite student fall guest girlfriend business convince spend
	Crime	murder suspect investigate death crime detective victim drug investigation killer
	—	challenge travel explore water history animal journey human country bear
	—	hospital daughter relationship offer husband continue death arrive deal sister
	—	beaver class arrive animal parent catch fall idea surprise teacher
	Fantasy	power king magic fight evil demon battle attack spell human
	Game-Show	contestant challenge answer final round question prize winner judge compete
	History	king history country century power queen empire battle french army
	Horror	dean ghost spirit town zombie haunt death vampire paranormal horror
	Music	perform song dance band sing musical guest performance update rock
	—	wedding dress relationship business lady bride idea dream husband marry
	Crime	murder death investigate body victim suspect killer detective local crime
	News	news president program issue political campaign rebroadcast examine segment health
	Reality-TV	challenge season model create food crew fashion dream race travel
	Romance	accept wedding victor ridge offer hospital kiss convince arrive marriage
	Sci-Fi	planet alien ship space power doctor human crew destroy force
	Short	chef food cook restaurant dish recipe prepare meal menu kitchen
	Sport	player fight coach match race season football champion battle championship
	Talk-Show	guest actor director fashion book chat round table production panelist
	Crime	death secret investigate force murder continue reveal attack deal body
	War	army german soldier force officer colonel military battle troop attack
	Western	town horse ranch train gang ride force arrive wagon cattle